Windows Server Backup

Introduction

- Windows Server Backup (WSB) is a **built-in, free backup and recovery feature** in Windows Server
- It was introduced in Windows Server 2008 and replaced the older NTBACKUP tool.
- WSB provides **wizards and command-line tools** for performing basic backup and recovery tasks
- **Installation**
  - **GUI Method**
    - Open **Server Manager** → **Add Roles and Features**
    - Select **Features** (not Roles)
    - Check **Windows Server Backup**
    - Click **Install**
  - **PowerShell Method**

```
Install-WindowsFeature -Name Windows-Server-Backup
```

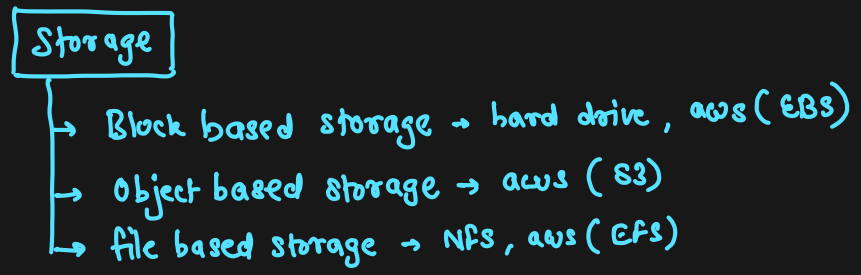
Backup Types

- **Full Server:** All volumes, including OS, system state, and all data
- **Selected Volumes:** Only specified volumes
- **System State:** Critical system components (see Section 4)
- **Specific Files/Folders:** Fine-grained file-level backup
- **Bare Metal Recovery (BMR):** Entire system, for disaster recovery to new hardware

Technical Features

↳ copy the volume block by block

- **Block-based Backup:** WSB integrates with Volume Shadow Copy Service (VSS) and works at the block level, enabling snapshot-style backups. It can back up any application with a VSS Writer
- **Incremental and Full Backups:** Supports incremental backups (prioritizes backup speed) and full backups (prioritizes post-recovery I/O performance, does not retain shadow copies on the volume)
- **Automatic Disk Usage Management:** Configurable deletion policies automatically remove older backups to optimize disk space
- **Large Volume Support:** Supports volumes larger than 2 TB and 4K sector sizes
- **Cluster Shared Volume (CSV) Support:** Limited CSV backup support (VMs cannot be added as part of the backup configuration; volume recovery is not supported)
- **Hyper-V Support:** Full support for backing up and recovering Hyper-V hosts and virtual machines.



System State

- System State is a collection of critical components
- You cannot choose individual components because of dependencies among them
  - Boot files (including system files protected by Windows File Protection)
  - COM+ Class Registration Database
  - Active Directory (on Domain Controllers only)
  - SYSVOL (on Domain Controllers only)
  - Certificate Services (on Certificate Authority only)
  - Cluster database (on cluster nodes only)
  - IIS metabase (on IIS servers only)
  - Performance counter configuration
  - Registry
- **System State Limitations**
  - Cannot back up to a critical volume
  - Cannot be used for BMR alone (requires full server backup)
  - Size varies by server role

Backup Destinations

- Dedicated Local Disk
  - A disk dedicated exclusively to backups
  - Format is handled automatically by WSB
  - Recommended for scheduleackups
- Local Volume (Non-critical)
  - Can be a separate volume on the same disk
  - Must not contain the OS or system state
- Removable Media (USB/eSATA)
  - External drives can be used
  - Must be formatted appropriately
  - Suitable for offsite rotation
- Network Share (Remote SMB Share)
  - Backups can be written to a remote SMB share
  - Requires proper credentials
  - Limitations apply for system state / BMR backups
- DVD/CD (Legacy)
  - Not supported for scheduled backups
  - Only for manual, one-time backups (deprecated in newer versions)

How to Back Up System State

- Open **Windows Server Backup** (via Server Manager → Tools)
- Select **Local Backup** → **Backup Once**
- Choose **Different Options** → **Custom**
- Select **Add Items** → **System State** → OK
- Choose destination (Local drive or Remote shared folder)
- Complete the wizard

Recovery Capabilities

- Volume-level recovery
- File/folder-level recovery
- Application recovery (e.g., Exchange, SQL Server)
- System State recovery
- **Bare Metal Recovery (BMR)**: Restores a complete system to the original or new hardware

Recovery Operations



- **Recovery Types**
  - Files and Folders: Restore individual items
  - Volumes: Restore entire volumes
  - Applications: Restore app data (VSS-aware apps)
  - System State: Restore system components
  - Bare Metal Recovery: Full system to bare metal (machine)
- **Bare Metal Recovery (BMR) Process**
  - Boot from Windows Server installation media
  - Select Repair your computer
  - Choose System Image Recovery
  - Select the backup to restore
  - Start recovery